SURPAC Surveying Software

SURPAC Surveying Software consists of eight Modules, whose applications are accessed via seven pull-down Menus.

These eight Modules are :-

The SURPAC "Lite" Module consisting of the General Menu applications, plus certain applications fro other Menus.

The Conversions Module consisting of the Conversions Menu applications.

The Least Squares Module consisting of the Least Squares Menu applications.

The Topographical Menu consisting of the Topographical Menu applications.

The Engineering Module consisting of the Engineering Menu applications.

The Mining Module consisting of the Mining Menu applications.

The Cadastral (Standard) Module consisting of the top portion of the Cadastral Menu applications.

The Cadastral (Extended) Module consisting of the bottom portion of the Cadastral Menu applications

SURPAC Surveying Software							
SURPAC Module Structure							
SURPAC"Lite"	Conversions	Least Squares	Topographical	Engineering	Mining	Cadastral Standard	Cadastral Extended

The colour coding of the above table is used throughout these Web Pages to indicate the Module to which an individual application is allocated.

The SURPAC "Lite" Module is the only "stand alone" Module, and represents the minimum software configuration.

Any of the other seven Modules may be added to the SURPAC "Lite" Module , as per Users' requirements.

1. The SURPAC "Lite" Module

- The SURPAC "Lite" Module provides a Stand Alone, Entry Level suite of Survey Software applications, that covers the fundamental calculation and CAD requirements for most Survey Offices.

- The omission of specific applications found in the SURPAC Conversions, Least Squares, Topographical, Engineering, Mining and Cadastral Modules makes the SURPAC "Lite" Module a cost effective "starter" suite of Survey applications.

- In spite of these cost reducing omissions, the SURPAC "Lite" Module is able to provide a suite of Survey calculation solutions, as described below. Click the Button above in order to view a detailed description of the applications included in the SURPAC "Lite" Module.

- Purchasers of the SURPAC "Lite" Module may expand their SURPAC Software at any time, by adding on any of the other seven Modules. This may be done simultaneously, or at a future date.

- The SURPAC "Lite" Module is currently priced at R 5 500.00 in Southern Africa, and at US$ 960.00 for countries outside this region.

The SURPAC "Lite" Module includes the following applications :-

- Support for various South African and U.K. Survey Systems such as :
    - South African "WG" Hartebeeshoek94 Datum,
    - Southern African "Lo" Cape Datum,
    - Southern African Arc 1950 Datum,
    - U. K. National Grid (OSGB36),
    - Irish National Grid,
    - U. K. National GPS Network (ETRS89),
    - Universal Transverse Mercator,
    - and others.
- Support for Ellipsoids such as :
    - Clarke 1880 (Modified),
    - Clarke 1866,
    - WGS 1984,
    - WGS 1972,
    - Airy 1830,
    - Airy 1830 (modified),
    - Bessel 1841,
    - and others.
- 2D and 3D Join calculations,
- Polar calculations from plane data and from field observations (auto distance reduction and orientation),
- Reverse Polar calculations,

- Two Sides & the Included Angle calculations,
- Intersections and Trilaterations,
- Curve Tangent Point and 3 Point Curve fitting,
- Area calculations with or without Co-ordinate checking,
- Fixed Area and Pivot Point Calculations
- Off-Line Calculations and Least Squares Line fitting,
- Splay Point calculations,
- Apex Points calculations,
- Data Traverses,
- Field Traverses (using Observation Files)
- Line Running and adjustment,
- Total Station and Logger Downloading and Uploading of Co-ordinates,
- File Printing and Setting Out Sheet construction,
- Importing and Exporting [Latitude, Longitude, Height] ASCII files,
- Importing and Exporting [Y (or E), X (or N), Height] ASCII files,
- Importing and Exporting [Y (or E), X (or N), Height] DXF format files,
- Importing and Exporting [Y (or E), X (or N), Height] data from a *Microsoft* Access Database,
- General 2-D CAD drawing facility using Lines, Arcs, Text, Mark Points, Contours, plus DXF and Image support,
- Basic Co-ordinate, Distance and Area Conversions (from the Conversions Menu),
- Geographical Transformations : Ellipsoid to/from Plane and Panel to Panel (Conversions Menu),
- Standard Helmert (1st Order) System to System Transformations (Conversions Menu),
- Locating and Meaning Point Groups in a Co-ordinate File (Conversions Menu),
- Point comparisons using different Co-ordinate Files (Conversions Menu),
- [Y, X, Z] (or [E, N, H]) Resections and Intersections using Least Squares (Least Squares Menu),
- Direct and/or ASCII loading of Observations from a wide range of Total Stations and Loggers (Topographical Menu),
- Mass Polar reductions direct from Observation Files (Topographical Menu),
- Viewing ASCII Files and Databases,
- On-screen Calculator with Survey, mathematical and financial functions.

2. The SURPAC Conversions Module

The SURPAC Conversions Module includes the following applications :-

- Basic Co-ordinate, Distance and Area Conversions,
- Geographical Transformations : Ellipsoid to/from [Y, X] Plane System,
- Geographical Transformations : Ellipsoid to Ellipsoid,
- Geographical Transformations : System Panel 1 to System Panel 2,
- Standard Helmert (1st Order) System to System Transformations,
- Weighted Helmert (1st Order) System to System Transformations,
- Polynomial (2nd Order) System to System Transformations,
- Transformations from "Lo" RSA Cape Datum to "WG" Hartebeeshoek94 Datum,
- Transformations from "WG" Hartebeeshoek94 Datum to "Lo" RSA Cape Datum,
- RSA "GoldFields" System to/from "Lo" RSA Cape Datum,
- Locating and Importing nearest Trig./TSM control Points (RSA only),
- [DY, DX, DZ] stability precision Monitoring facility using multiple epochs,
- Locating and Meaning Point Groups in a Co-ordinate File,
- Point comparisons using different Co-ordinate Files.

3. The SURPAC Least Squares Module

The SURPAC Least Squares Module includes the following applications :-

- Point [Y, X, Z] determination for Resections, Intersections or Trilaterations,
- [Y, X] Planimetric Control Network Adjustment for up to 400 Points,
- Trig Height Network Adjustment for up to 500 Points,
- Spirit Level Network Adjustment for up to 500 Points.

4. The SURPAC Topographical Module

The SURPAC Topographical Module includes the following applications :-

- Direct and/or ASCII loading of Observations from a wide range of Instruments,
- Manual Observation Data entry and editing,
- Direct and/or ASCII loading of Spirit Level Observations from Instruments,
- Manual Spirit Level Observation Data entry and editing,
- Rapid Observation File reductions for Co-ordinate and Tacheometric Files,
- Tacheometric File Creation/Editing, plus importing/Exporting between ASCII and Co-ordinate Files,
- Contour Development CAD facility, using Surface Triangles and/or Break Line methods,
- Digital Terrain Modelling - 3D displays, Volumes, Areas, Sections and DTM comparisons.

5. The SURPAC Engineering Module

The **SURPAC Engineering Module** includes the following applications :-

- Horizontal Alignment (Straights, Circular & Transition Curves)
- Vertical Curve Alignment and Profile Formations
- Cross Section Creating and Plotting
- Longitudinal Section Creating and Plotting
- Sectional Volumes and Toe Pegs

6. The SURPAC Mining Module

The **SURPAC Mining Module** includes the following applications :-

- Peg Calculating and Checking using the Double Set Up Method,
- Peg Calculating and (Semi) Checking using the Double Button Method,
- Peg Calculating and (Semi) Checking using Double Back Fix Method,
- Viewing and re-using Peg ASCII files for Peg re-Calculations,
- Offset Surveys (Bord and Pillar) with manual or ASCII input,
- Offset Surveys using an Observation File,
- Pillar Safety Factor Calculations,
- Gyro-Theodolite Calibrations (using Transit Method or Schuler Means),
- Gyro-Theodolite Bearing Reductions (using Transit Method or Schuler Means).

7. The SURPAC Cadastral (Standard) Module

The **SURPAC Cadastral (Standard) Module** includes the following applications :-

- Diagram CAD facility for A0 - A4 Diagram Construction, Editing and Plotting,
- Working Plan CAD facility for Working Plan Construction, Editing and Plotting.

8. The SURPAC Cadastral (Extended) Module

The SURPAC Cadastral (Extended) Module includes the following applications :-

- General Plan CAD facility for GP Constructing, Editing and Plotting,
- GP Data Sheet CAD facility for DS Constructing, Editing and Plotting,
- Sectional Title Plan CAD facility for ST Constructing, Editing and Plotting.

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